

Jie He

Ph.D. Student in Computer Science at University of Edinburgh

• j.he@ed.ac.uk • [Website](#) • [LinkedIn](#) • [GitHub](#)

Research Interests

My research focuses on **LLM post-training**, primarily from four core perspectives: **Augmentation**, **Reasoning / Agents**, **Efficiency**, and **Multimodality**.

- **Augmentation:** Investigating the role of *knowledge* in the era of large language models, and how external knowledge sources can effectively complement parametric knowledge. [1, 3, 5, 8, 10]
- **Reasoning / Agents:** Enhancing LLMs' ability to leverage external knowledge and novel tools through *supervised fine-tuning (SFT)* and *reinforcement learning*. [1, 3, 4]
- **Efficiency:** Designing compression methods that reduce the length of external inputs, enabling models to improve efficiency without sacrificing performance. [3]
- **Multimodality:** Studying efficient training strategies for multimodal LLMs and analyzing their reasoning deficiencies. [7, 9]

In addition, I am interested in **LLM evaluation and benchmark construction**, particularly in designing challenging scenarios for rigorous assessment of question answering and reasoning. [2, 6, 7, 13, 14, 15]

Furthermore, I proposed **CLARA**, one of the earliest open-source frameworks that jointly learns retrieval and reasoning through latent vectors. The project has received **1100+ stars on GitHub**.

Education

School of Informatics, University of Edinburgh

Ph.D. in ILCC

Advisor: [Prof. Jeff Z. Pan](#)

2022 – Present

Edinburgh, UK

Department of Computer Science, Tianjin University

M.S. in Computer Science

Advisor: [Prof. Deyi Xiong](#)

2019 – 2022

Tianjin, China

Department of Software Engineering, Shandong University

B.S. in Software Engineering

2015 – 2019

Shandong, China

Work Experience

NVIDIA

Research Intern @ NVIDIA Nemo Retriever Team

Project: Train a SOTA embedding model for reasoning agentic retrieval.

Model: [llama-nv-embed-reasoning-3b](#)

Mentor: [Yauhen Babakhin](#), [Dr. Ronay Ak](#)

2025.10 – 2026.02

Remote

APPLE

Research Intern @ APPLE MLR

2025.06 – 2025.09

Seattle, US

Project: Improve the retrieval quality and generation efficiency of LLM with latent vectors.

Mentor: [Dr. Yizhe Zhang](#)

MICROSOFT

2024.07 – 2024.10

Research Intern @ OAR & MSR

Redmond, US

Project: Enhancing Tool Generalization in LLMs.

Mentor: [Dr. Pei Zhou](#), [Dr. Longqi Yang](#), [Prof. Jennifer Neville](#)

APPLE

2024.02 – 2024.05

Research Intern @ APPLE AI/ML

Seattle, US

Project: Investigating to generate rationales for knowledge-based question answering without training.

Mentor: [Dr. Yiwen Sun](#), [Dr. Benjamin Han](#)

Honors and Awards

- *Apple Scholars in AI/ML* PhD fellowship 2024 – 2026
- *PhD Scholarship*, School of Informatics, University of Edinburgh 10/2022 – 09/2025

Publications

“*” indicates equal contribution

1. **From Reflection to Sufficiency: Reinforcement-Guided Thinking Quality in Retrieval-Augmented Reasoning for LLMs.**
Jie He, Victor Gutierrez Basulto, Jeff Z. Pan.
EMNLP 2026.
2. **Clara: Bridging Retrieval and Generation with Continuous Latent Reasoning.**
Jie He, Richard He Bai, Sinead Williamson, Jeff Z. Pan, Navdeep Jaitly, Yizhe Zhang.
COLM 2026.
3. **MINTQA: A Multi-Hop Question Answering Benchmark for Evaluating LLMs on New and Tail Knowledge.**
Jie He*, Nan Hu*, Wanqiu Long, Jiaoyan Chen, Jeff Z. Pan.
ACL 2026.
4. **GenTool: Enhancing Tool Generalization in Language Models through Zero-to-One and Weak-to-Strong Simulation.**
Jie He, Jennifer Neville, Mengting Wan, Longqi Yang, Hui Liu, Xiaofeng Xu, Xia Song, Jeff Z. Pan, Pei Zhou.
Findings of ACL 2025.
5. **kNN-XICL: Cross-lingual In-Context Learning with Nearest Neighbor Inference.**
Jie He, Simon Yu, Victor Gutierrez Basulto, Jeff Z. Pan.
IJCNN Workshop 2025.
6. **Exploring Knowledge Graph to Text Generation with Large Language Models: Techniques, Challenges, and Innovations.**
Jie He*, Yijun Yang*, Wanqiu Long, Victor Gutierrez Basulto, Jeff Z. Pan.
NAACL 2025.

7. **MiCEval: Unveiling Multimodal Chain of Thought’s Quality via Image Description and Reasoning Steps.**
Xionghao Zhou*, **Jie He***, Lanyu Chen, Jingyu Li, Haojing Chen, Victor Gutierrez Basulto, Jeff Z. Pan, Hanjie Chen.
NAACL 2025.
8. **Evaluating the Adversarial Robustness of Retrieval-Based In-Context Learning for Large Language Models.**
Simon Yu*, **Jie He***, Pasquale Minervini, Jeff Z. Pan.
COLM 2024.
9. **An Empirical Study on Parameter-Efficient Fine-Tuning for Multimodal Large Language Models.**
Xionghao Zhou*, **Jie He***, Yuhua Ke, Guangyao Zhu, Victor Gutierrez Basulto, Jeff Z. Pan.
Findings of ACL 2024.
10. **BUCA: A Binary Classification Approach to Unsupervised Commonsense Question Answering.**
Jie He, Simon Yu, Victor Gutierrez Basulto, Jeff Z. Pan.
ACL 2023.
11. **Instances and Labels: Hierarchy-aware Joint Supervised Contrastive Learning for Hierarchical Multi-Label Text Classification.**
Simon Yu*, **Jie He***, Victor Gutierrez Basulto, Jeff Z. Pan.
Findings of EMNLP 2023.
12. **MetaXCR: Reinforcement-Based Meta-Transfer Learning for Cross-Lingual Commonsense Reasoning.**
Jie He, Yu Fu.
NeurIPS Workshop 2022.
13. **Evaluate Discourse Cohesion in Pre-trained Language Models.**
Jie He, Wanqiu Long, Deyi Xiong.
COLING Workshop 2022.
14. **TGError: An Error-Annotated Dataset and Benchmark Tasks for Text Generation from Pretrained Language Models.**
Jie He*, Bo Peng*, Yi Liao, Qun Liu, Deyi Xiong.
ACL 2021.
15. **The Box is in the Pen: Evaluating Commonsense Reasoning in Neural Machine Translation.**
Jie He*, Tao Wang*, Deyi Xiong, Qun Liu.
Findings of EMNLP 2020.
16. **UniArk: A Holistic Approach to Unbiased and Consistent Factual Knowledge Extraction.**
Yijun Yang, **Jie He**, Pinzhen Chen, Victor Gutierrez Basulto, Jeff Z. Pan.
NAACL 2024.
17. **Similarity-Based Domain Adaptation with LLMs for Cross-Domain Sentiment Classification** [\[Preprint\]](#).
Jie He, Wendi Zhou, Xiang Lorraine Li, Jeff Z. Pan.

18. **Meta-RTL: Reinforcement-Based Meta-Transfer Learning for Low-Resource Commonsense Reasoning** [\[Preprint\]](#).

Yu Fu*, **Jie He***, Yifang Yang, Qun Liu, Deyi Xiong.

Academic Services

- **Reviewers:** *AACL (2022, 2023)*, *SIGDIAL (2023)*, *EMNLP (2023)*, *EACL (2024)*, *NAACL (2024)*, *ACL (2024)*, *ARR (2023.10 – now)*, *COLM (2024)*, *ICLR (2025)*, *COLING (2025)*
- **Area Chair:** *EMNLP (2024)*, *ARR (2024.06 – now)*

Teaching Experience

Knowledge Graph, University of Edinburgh

Fall 2023

Tutor